Homework 11 due Nov 19 before class

1. A light of intensity I_0 falls on a layer of plasma (a mix of free ions and electrons) with density n (in units cm^{-3}, assume single charged) and thickness l (cm). Find intensity of light that passed upscattered (assume heavy ions do not respond to radiation, and wave-electron interaction occurs in single-particle regime).

2. Why a simple charge in circular motion emits, while constant loop current does not?

3. A dielectric sphere of radius R with dielectric constant ϵ has constant charge density ρ_e. Find electric field (both inside and outside).

   Hint:

$$\begin{aligned} \operatorname{div} \mathbf{D} &= 4\pi\rho_e \\ \mathbf{E} &= -\nabla\Phi \\ \mathbf{D} &= \epsilon\mathbf{E} \end{aligned} \tag{1}$$

and the boundary condition is

$$\epsilon\partial_r\Phi|_{R_-} = \partial_r\Phi|_{R_+} \tag{2}$$